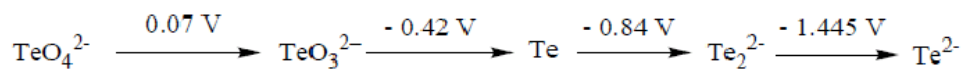


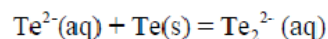
8. (a) Use the Latimer diagram below to construct a Frost diagram for tellurium in basic solution, pH = 14.



[6]

- (b) Calculate the following

- The standard potential for the $\text{TeO}_4^{2-} / \text{Te}$ couple at pH=14.
- The equilibrium constant for the reaction at pH=14.



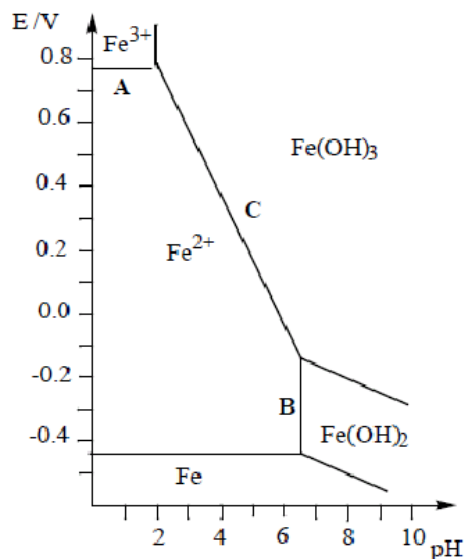
[5]

- (c) A simplified Pourbaix diagram for iron is given below.

- Write equations for the equilibria described by the boundary lines A, B, and C.
- Account for the slope of line C.
- Deduce, using the following data (all at pH=0), what iron species are expected to be present in oxygenated water at pH=7.

$$E^\circ (\text{H}^+ / \text{H}_2) = 0 \text{ V} \quad E^\circ (\text{O}_2 / \text{H}_2\text{O}) = 1.229 \text{ V}$$

$$[RT/F = 0.026 \text{ V}; (RT/F)\log_{10}e = 0.059 \text{ V at } 298 \text{ K}]$$



[14]